Riemannian and Complex Geometry

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Submission deadline: 20 December 2025, 5:59 pm (GMT-4)

— Exercises —

6.1 Hermitian metric

Exercise 6.1. Let (E,h) be a Hermitian vector bundle over a complex manifold X. A connection ∇ is a metric connection if and only if

$$\mathrm{d}\{s,t\} = \{\nabla s, t\} + (-1)^p\{s, \nabla t\},$$

where $s \in C^\infty(X, \Omega^p_{X,\mathbb{C}} \otimes E)$ and $t \in C^\infty(X, \Omega^q_{X,\mathbb{C}} \otimes E)$.

Exercise 6.2. Let (E,h) be a Hermitian vector bundle equipped with connection ∇^E. Then ∇^E is a metric connection if and only if $\nabla^{E^*\otimes\overline{E}^*}h = 0$.

Exercise 6.3. Let $\mathbb{CP}^n = \bigcup_{i=0}^n U_i$ be the canonical open covering, that is $U_i = \{(z^0 : \cdots : z^n) \mid z^i \neq 0\}$. Then there is a Hermitian metric h on the holomorphic tangent bundle of $\mathbb{CP}^n$, called Fubini-Study metric, such that

$$\omega|_{U_i} = \frac{\sqrt{-1}}{2}\partial\overline{\partial}\log\left(\frac{\sum_{l=0}^n |z^\ell|^2}{|z^i|^2}\right),$$

where $\omega = h_{i\overline{j}}\mathrm{d}z^i \wedge \mathrm{d}\overline{z}^j$.

6.2 Chern connection and Chern curvature

Exercise 6.4. Show that the curvature of constant hermitian metric on tangent bundle of a complex torus $\mathbb{C}^n/\Gamma$ is trivial. Is this true for any hermitian metric on tangent bundle of $\mathbb{C}^n/\Gamma$.

Exercise 6.5. Let (E,h) be a Hermitian holomorphic vector bundle on a complex manifold X and ∇ be Chern connection. For $s \in C^\infty(X,E)$, locally written as $s = s^\alpha e_\alpha$, prove that

$$\nabla_{\frac{\partial}{\partial z^i}}\nabla_{\frac{\partial}{\partial \overline{z}^j}} s^\beta - \nabla_{\frac{\partial}{\partial \overline{z}^j}}\nabla_{\frac{\partial}{\partial z^i}} s^\beta = \Theta^\beta_{i\overline{j}\alpha}s^\alpha.$$

Exercise 6.6. Let (E,h) be a Hermitian holomorphic vector bundle of rank r over a complex n-manifold X with Chern connection ∇.

(1) If (E,h) is Nakano positive or dual Nakano positive, then (E,h) is Griffiths positive.

(2) (E,h) is Nakano positive if and only if (E^*,h^*) is dual Nakano negative.

6.3 Chern class

Exercise 6.7. Prove the following formula by definition

$$c_1(E \otimes F) = \mathrm{rk}(F)c_1(E) + \mathrm{rk}(E)c_1(F),$$

where E, F are complex vector bundles on a complex manifold.

Exercise 6.8. Compute

$$\int_{\mathbb{CP}^n} c_1(T\,\mathbb{CP}^n)^n$$

by Fubini-Study metric on $\mathbb{CP}^n$.